iNeuBytes – Task 1

Airline Loyalty Data Analytics Using SQL

1. Project Overview

This project focuses on analyzing an airline loyalty program dataset using SQL and SQLite. The objective is to understand customer behavior, flight activity, loyalty program performance, and Customer Lifetime Value (CLV).

2. Tools Used

- DB Browser for SQLite
- SQL
- SQLite Database

3. Database Details

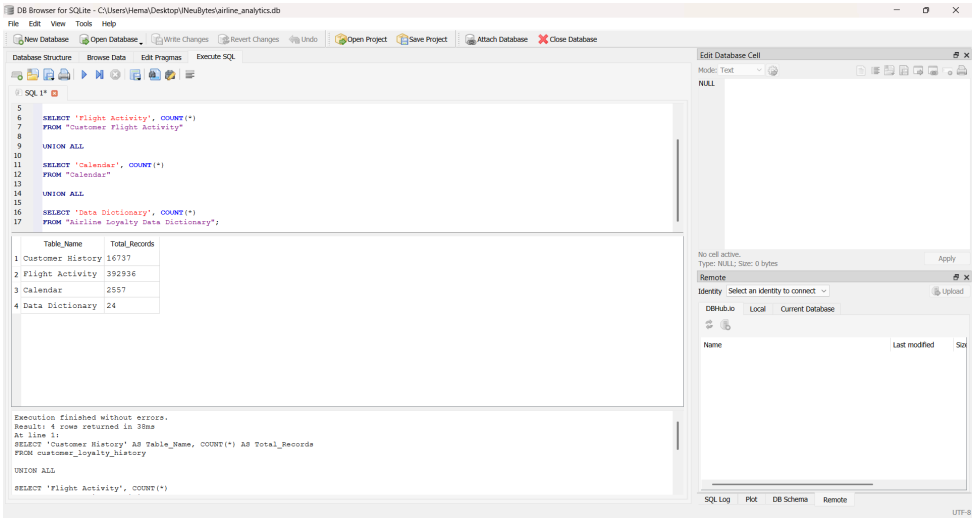
Database name: `airline_analytics.db`

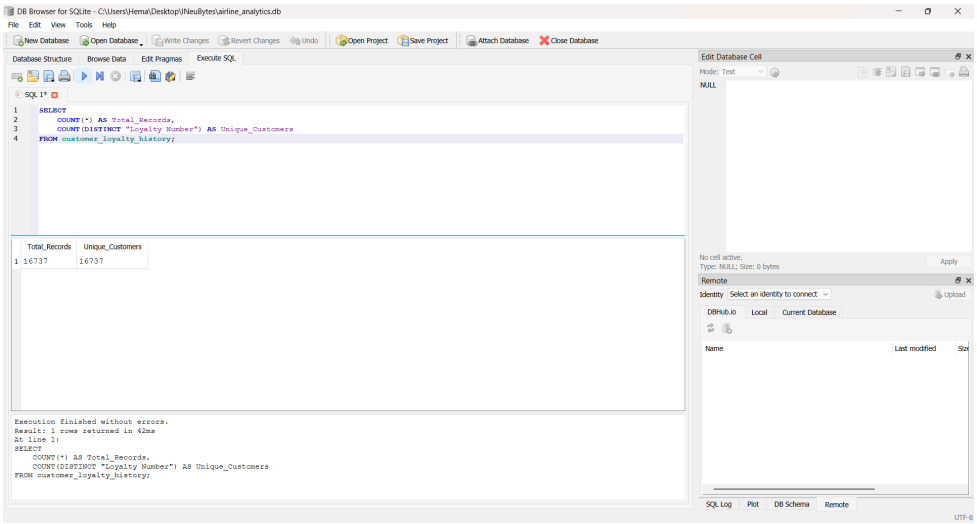
The database contains four tables:

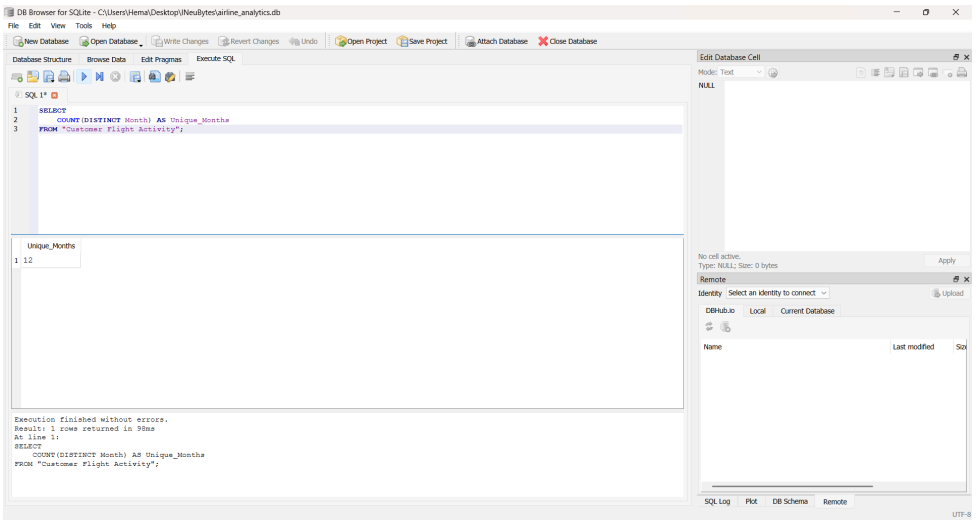
1. `customer_loyalty_history`
2. `Customer Flight Activity`
3. `Calendar`
4. `Airline Loyalty Data Dictionary`

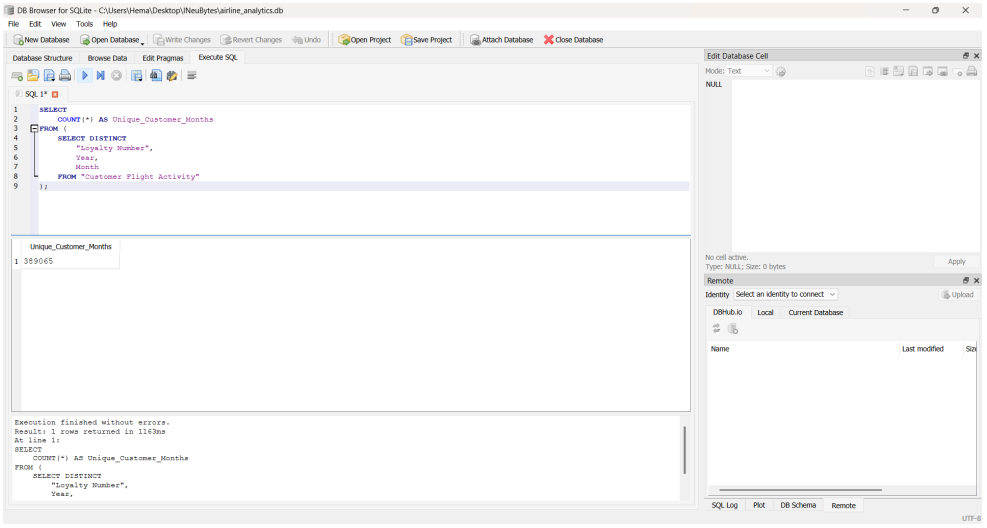
4. Dataset Overview

Metric	Result
Customer history records	16,737
Unique customers	16,737
Flight activity records	392,936
Calendar records	2,557
Data dictionary records	24
Unique months in flight activity	24
Unique customer-month combinations	389,065


DB Browser for SQLite - C:\Users\Hema\Desktop\NeuBytes\airline_analytics.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
SQL 1*
SELECT 'Flight Activity', COUNT(*)
FROM "Customer Flight Activity"
UNION ALL
SELECT 'Calendar', COUNT(*)
FROM "Calendar"
UNION ALL
SELECT 'Data Dictionary', COUNT(*)
FROM "Airline Loyalty Data Dictionary";
Table_Name Total_Records
1 Customer History 16737
2 Flight Activity 392936
3 Calendar 2557
4 Data Dictionary 24
Execution finished without errors.
Result: 4 rows returned in 38ms
At line 1:
SELECT 'Customer History' AS Table_Name, COUNT(*) AS Total_Records
FROM customer_loyalty_history
UNION ALL
SELECT 'Flight Activity', COUNT(*)
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified
SQL Log Plot DB Schema Remote
UTF-8


DB Browser for SQLite - C:\Users\Hema\Desktop\NeuBytes\airline_analytics.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
SQL 1*
SELECT
COUNT(*) AS Total_Records,
COUNT(DISTINCT "Loyalty Number") AS Unique_Customers
FROM customer_loyalty_history;
Total_Records Unique_Customers
1 16737 16737
Execution finished without errors.
Result: 1 rows returned in 42ms
At line 1:
SELECT
COUNT(*) AS Total_Records,
COUNT(DISTINCT "Loyalty Number") AS Unique_Customers
FROM customer_loyalty_history;
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified
SQL Log Plot DB Schema Remote
UTF-8


DB Browser for SQLite - C:\Users\Hema\Desktop\NeuBytes\airline_analytics.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
SQL 1*
SELECT
COUNT(DISTINCT Month) AS Unique_Months
FROM "Customer Flight Activity";
Unique_Months
1 12
Execution finished without errors.
Result: 1 rows returned in 98ms
At line 1:
SELECT
COUNT(DISTINCT Month) AS Unique_Months
FROM "Customer Flight Activity";
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified
SQL Log Plot DB Schema Remote
UTF-8



5. Data Validation and Quality Checks

The following checks were performed:

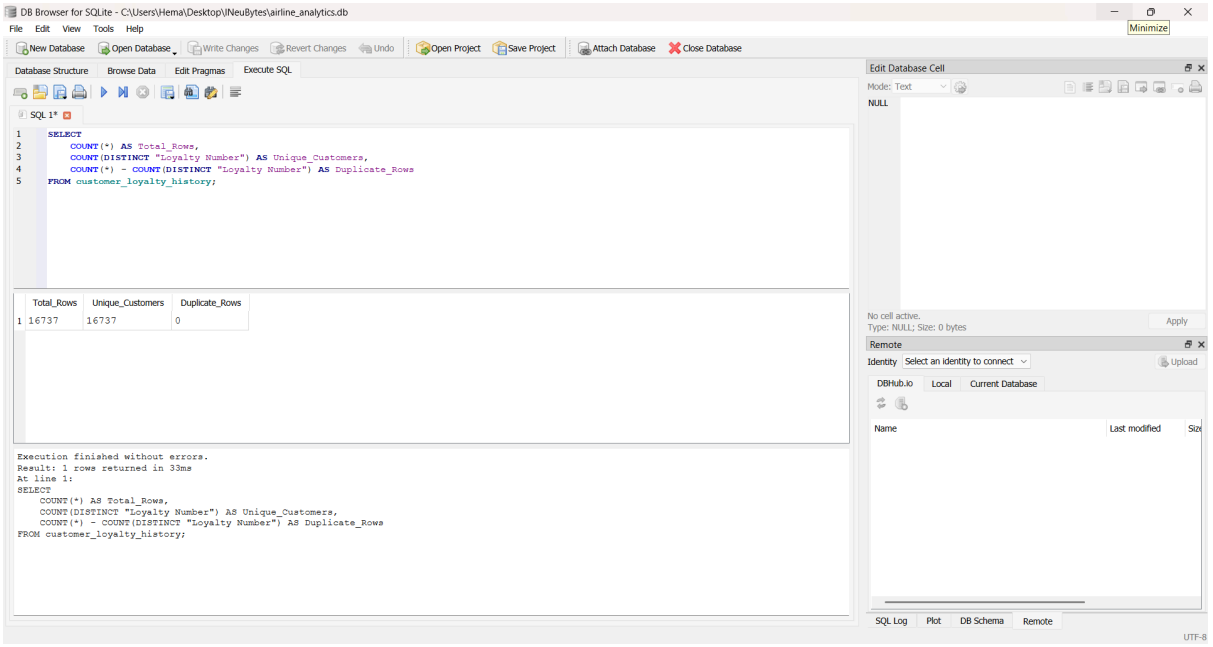
- Inspected table structures and column names.
- Checked missing values.
- Checked duplicate customer records.
- Checked duplicate customer-month records.
- Analyzed zero-value flight activity.
- Validated the relationship between customer history and flight activity.

Customer history duplicate check:

Metric	Result
Total rows	16,737
Unique customers	16,737
Duplicate rows	0

The customer history table contains no duplicate customer records.

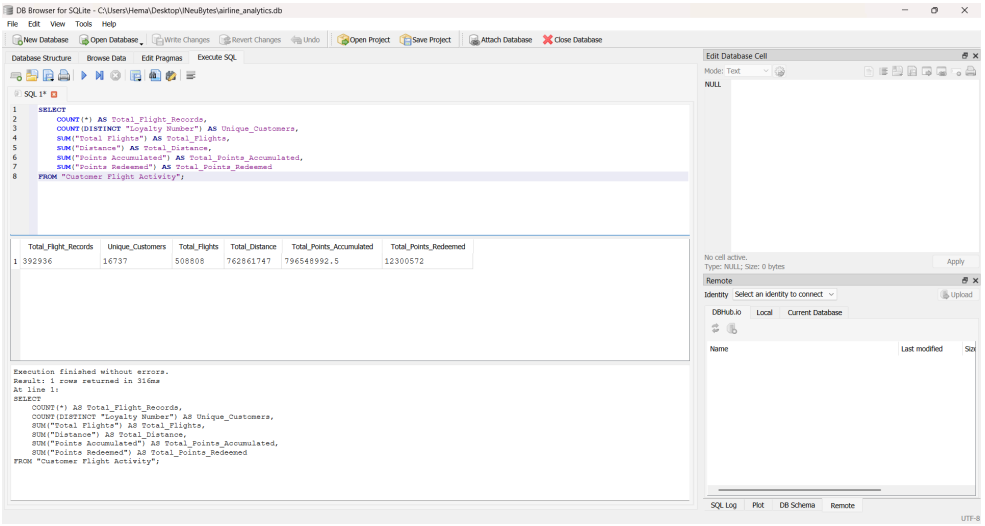
The flight activity and customer history tables also showed 392,936 matching records and zero unmatched records during JOIN validation.



6. SQL Analysis Performed

The following analyses were completed:

1. Overall flight performance
2. Year-wise flight performance
3. Month-wise flight performance
4. Loyalty Card-wise Customer Lifetime Value
5. Gender-wise Customer Lifetime Value
6. Education-wise Customer Lifetime Value
7. Country-wise customer analysis
8. Active vs Cancelled customer analysis
9. Enrollment Year analysis
10. Enrollment Type analysis
11. Points Redemption analysis
12. Flight Activity by Loyalty Card
13. Average Flight Activity per Customer
14. Customer Segmentation by CLV
15. Customer Cohort Retention Analysis

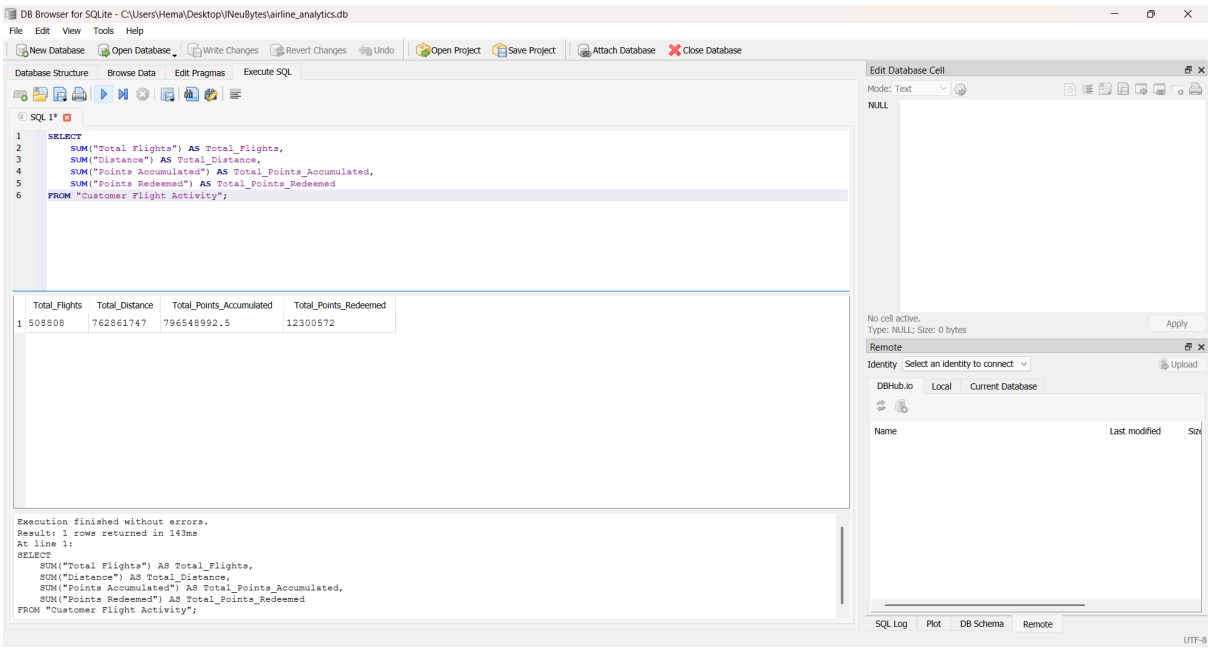


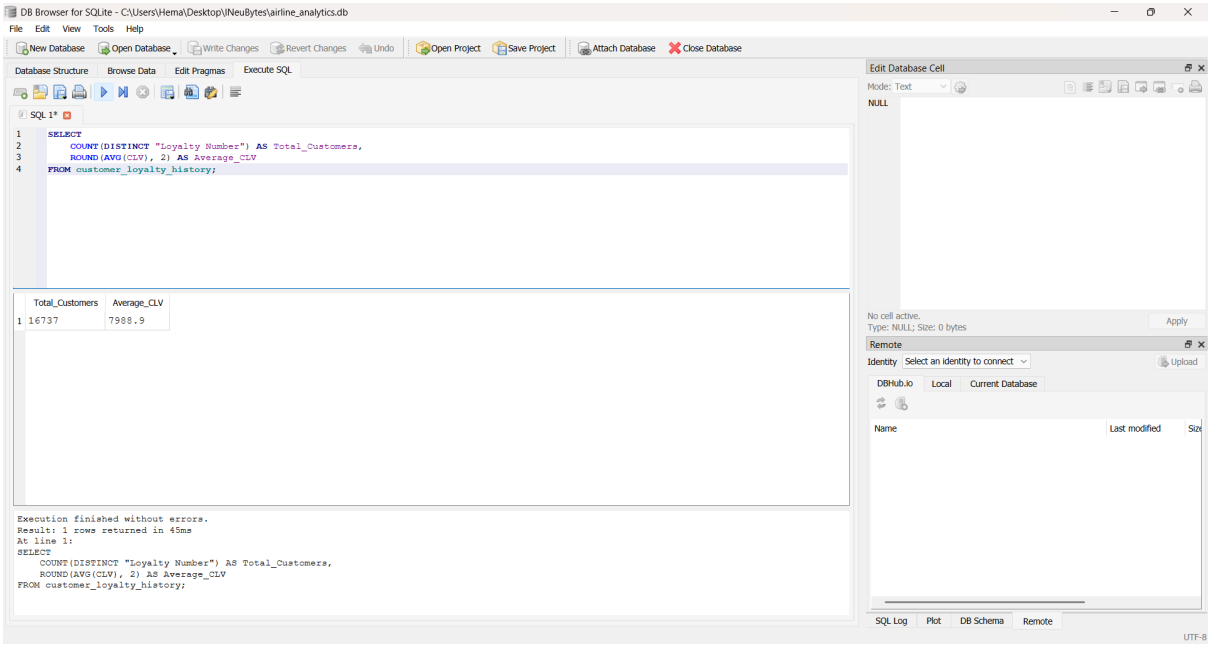
6.1 Key Performance Indicators (KPI) Analysis

The following six KPIs were calculated:

1. Total Flights
2. Total Distance
3. Total Points Accumulated
4. Total Points Redeemed
5. Total Customers
6. Average Customer Lifetime Value (CLV)

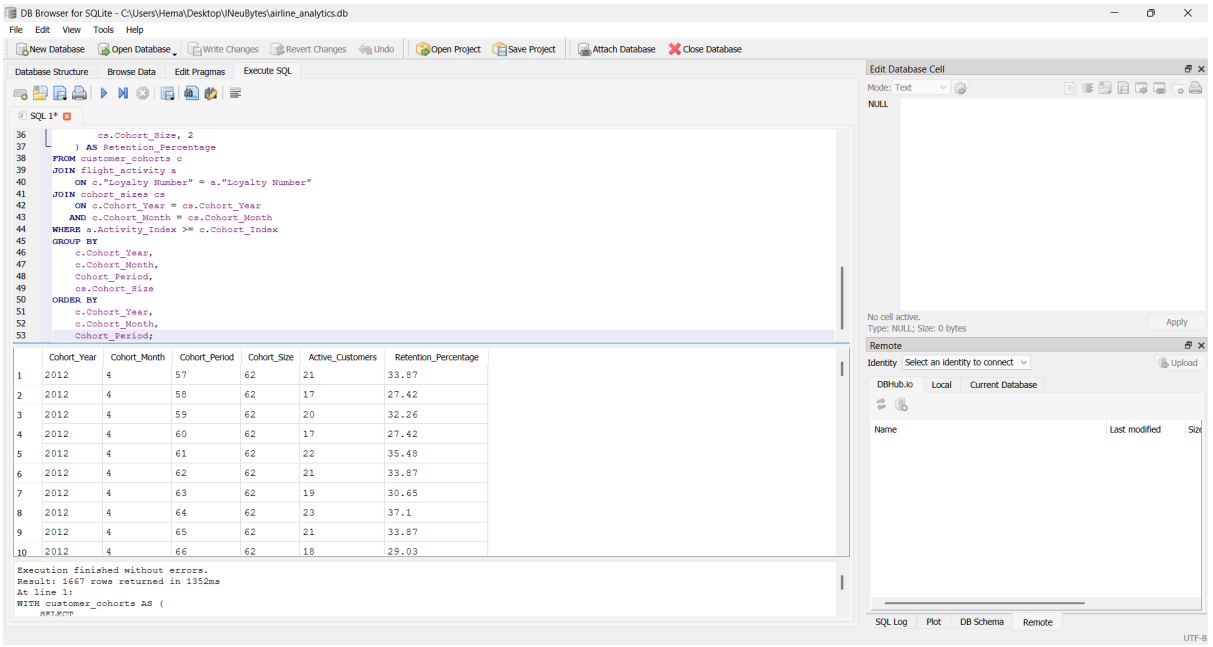
Observation: The analysis identified 16,737 unique customers, 508,808 total flights, and an average CLV of 7,988.90.





6.2 Customer Cohort Retention Analysis

SQL Query Output Screenshot



Explanation:

This analysis groups customers based on their enrollment year and month. It tracks customer flight activity in subsequent months and calculates the retention percentage to understand customer engagement and retention patterns.

Observation:

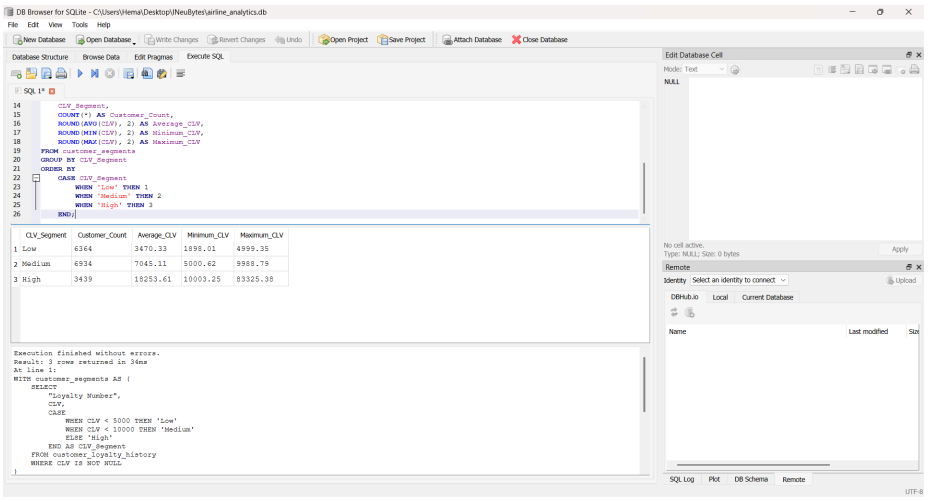
- The query returned 1,667 rows.

- Retention percentage varies across cohort periods.
- This analysis helps identify changes in customer retention over time

7. Customer Segmentation by CLV

Customers were classified into three segments using illustrative CLV thresholds.

Segment	CLV Range	Customer Count	Average CLV
Low	Below 5,000	6,364	3,470.33
Medium	5,000–9,999.99	6,934	7,045.11
High	10,000 and above	3,439	18,253.61
Total		**16,737**	



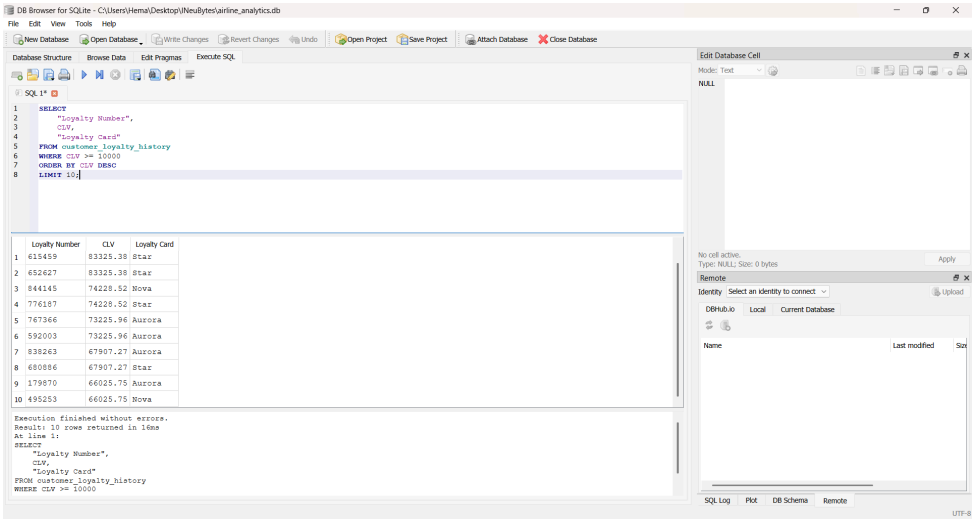
Observation:

- Medium-CLV customers form the largest segment.
- High-CLV customers represent an important group for customer retention.
- Low-CLV customers provide an opportunity for targeted engagement campaigns.

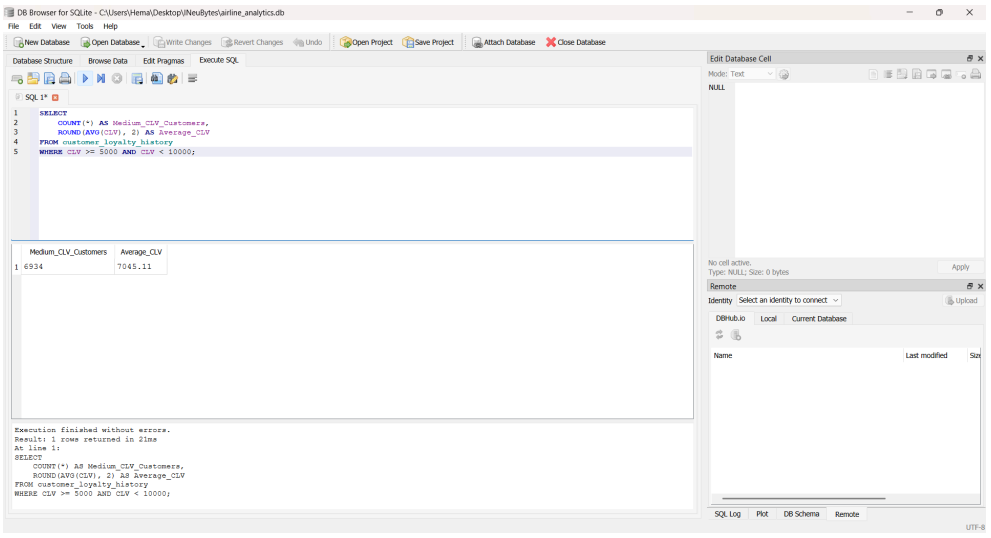
Note: These thresholds are illustrative business classifications, not official airline categories.

8. Key Business Recommendations

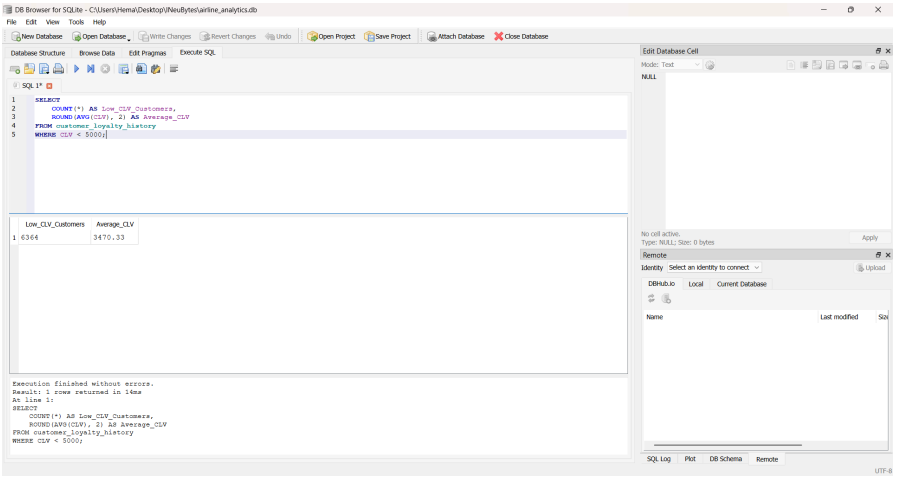
1. Provide exclusive rewards and premium benefits to high-CLV customers.



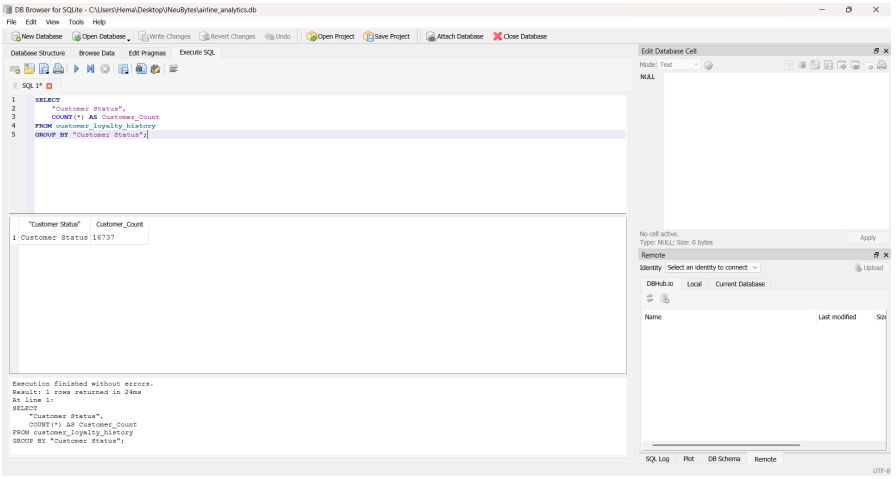
2. Encourage medium-CLV customers to increase their flight activity through targeted offers.



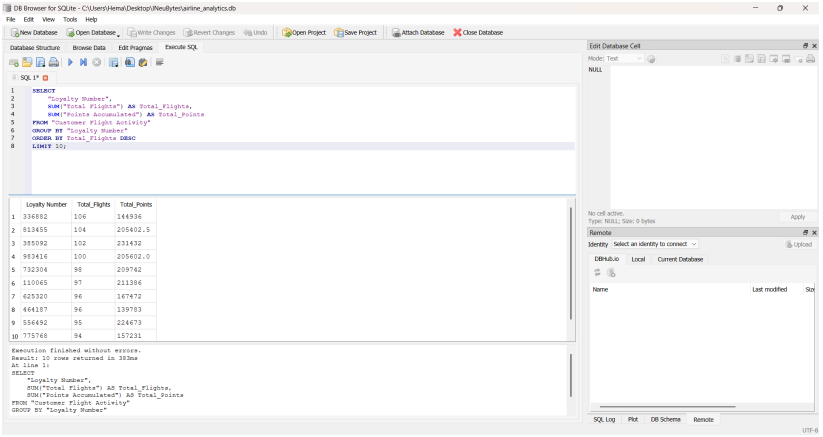
3. Create personalized campaigns to engage low-CLV customers.



4. Analyze cancellation patterns to improve customer retention.



5. Use customer activity trends to plan effective loyalty campaigns.



9. Project Conclusion

This project demonstrates how SQL can be used to analyze airline loyalty program data. By examining customer demographics, flight activity, loyalty card distribution, points redemption, and Customer Lifetime Value, the analysis helps identify customer behavior and valuable customer segments.

The findings can support data-driven decisions, improve customer engagement, strengthen customer retention, and enhance airline loyalty program strategies.

10. Skills Learned

- SQL queries and filtering
- Aggregate functions
- GROUP BY and ORDER BY
- JOIN operations
- Common Table Expressions (CTEs)
- Data cleaning and validation
- Customer segmentation
- Business insight generation

Project Status: SQL analysis completed. Final report prepared using the verified results available.